A Comparative Study of Machine Learning Models for Short-Term Stock Price Direction Prediction Using Public Market Data

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***Abstract*—Predicting the short-term direction of stock prices remains a challenging problem in computational finance due to the noisy and non-stationary nature of financial time series. This study presents a systematic comparative analysis of seven widely-used machine learning classification algorithms for predicting the 5-day price movement direction of stocks listed on the NIFTY 50 index of the National Stock Exchange of India. Using publicly available historical market data spanning January 2016 to December 2024, we construct a feature set of 31 technical indicators encompassing momentum, volatility, trend, and volume-based measures. The models evaluated include Logistic Regression, Support Vector Machines with radial basis function kernel, Random Forest, XGBoost, K-Nearest Neighbors, Gaussian Naive Bayes, and Gradient Boosting classifiers. We employ a temporal train-test split to preserve chronological ordering and avoid look-ahead bias. Our results indicate that ensemble tree-based methods, particularly Random Forest (mean accuracy 53.29%) and XGBoost (52.23%), marginally outperform linear and instance-based approaches across the selected stocks. The Friedman test confirms statistically significant differences among models ($\chi^2 = 15.36$, $p = 0.018$). Feature importance analysis reveals that trend strength indicators (ADX), volatility measures (ATR, Bollinger Band width), and MACD-derived features contribute most to predictive performance. However, the modest improvement over random chance across all models suggests that publicly available technical indicators alone provide limited predictive power for short-term price movements, consistent with the weak-form efficient market hypothesis.**

***Index Terms*—stock price prediction, machine learning, classification, technical indicators, NIFTY 50, random forest, XGBoost, financial time series**

I. Introduction

THE prediction of stock market movements has long attracted researchers and practitioners in finance, statistics, and more recently, machine learning. The ability to forecast even the directional movement of stock prices—whether prices will rise or fall over a given horizon—has significant implications for portfolio management, algorithmic trading, and risk assessment. Despite decades of research, this problem remains challenging due to the inherently noisy, non-linear, and non-stationary characteristics of financial time series data [1].

The Efficient Market Hypothesis (EMH) proposed by Fama [1] posits that asset prices fully reflect all available information, implying that consistent prediction of future price movements from historical data should be impossible in its weak form. However, a substantial body of empirical work has documented anomalies and patterns that machine learning techniques might exploit [3], [4]. This tension between market efficiency theory and empirical predictability motivates continued investigation into whether modern computational methods can extract meaningful signals from public market data.

Machine learning approaches have gained prominence in financial prediction due to their ability to model complex non-linear relationships without requiring explicit specification of the underlying data-generating process [5]. Various algorithms have been applied to this problem, ranging from classical statistical methods such as logistic regression to more sophisticated ensemble and kernel-based approaches. However, the comparative performance of these methods specifically on Indian equity markets, using a standardized experimental framework, remains insufficiently documented in the literature.

The Indian stock market, represented by the NIFTY 50 index on the National Stock Exchange (NSE), presents a particularly interesting testbed for several reasons. First, it is one of the largest and most liquid emerging markets, with characteristics distinct from the frequently-studied US and European markets [4]. Second, the market microstructure, including trading hours, settlement mechanisms, and participant composition, differs from developed markets in ways that may influence the efficacy of technical analysis-based features. Third, the availability of high-quality historical data through public APIs enables reproducible research.

This study makes the following contributions:

1) We present a rigorous comparative evaluation of seven machine learning classifiers for 5-day stock price direction prediction on 10 representative NIFTY 50 stocks spanning multiple sectors, using nine years of historical data (2016–2024).
2) We employ a strictly temporal train-test split that avoids the look-ahead bias commonly present in studies that use random cross-validation splits for time series data.
3) We conduct formal statistical significance testing using the Friedman test and post-hoc Wilcoxon signed-rank tests with Bonferroni correction to determine whether observed performance differences are genuine or attributable to chance.
4) We provide a detailed feature importance analysis identifying which categories of technical indicators contribute most to prediction accuracy, offering practical insights for feature engineering in financial machine learning applications.

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The remainder of this paper is organized as follows. Section II reviews related work in stock prediction using machine learning. Section III describes our data, features, models, and experimental methodology. Section IV presents the empirical results. Section V discusses findings and their implications. Section VI concludes the paper with directions for future research.

II. Related Work

A. Machine Learning in Stock Prediction

The application of machine learning to stock market prediction has evolved considerably over the past two decades. Early work by Huang et al. [2] applied Support Vector Machines (SVM) to predict the weekly movement direction of the NIKKEI 225 index, reporting accuracy rates of approximately 73%. However, subsequent attempts to replicate such results on different markets and time periods have generally yielded more modest outcomes [6].

Kara et al. [3] compared artificial neural networks (ANN) and SVM for predicting the direction of the Istanbul Stock Exchange national 100 index, using 10 technical indicators as input features. Their study reported ANN accuracy of 75.74% and SVM accuracy of 71.52%. However, these results were obtained using a random train-test split rather than a temporal one, potentially introducing look-ahead bias.

Patel et al. [4] conducted a comparative study of four machine learning algorithms—ANN, SVM, Random Forest, and Naive Bayes—on Indian stock market indices (CNX Nifty and S&P BSE Sensex) using technical indicators as features. They reported Random Forest as the best performer with accuracy exceeding 83% when using trend-based features derived from indicators. However, their approach of discretizing continuous indicators into binary trend values may limit generalizability.

Ballings et al. [6] performed a large-scale comparison of ensemble methods for stock price direction prediction across 5767 European stocks. Their study found that Random Forest outperformed other methods including neural networks, logistic regression, and SVM, with AUC values around 0.55–0.60 on out-of-sample data when using fundamental and technical features jointly.

B. Ensemble Methods in Finance

Ensemble methods have attracted particular attention in financial prediction. Gradient boosting variants, including XGBoost [7], have shown competitive performance in various financial prediction tasks. Krauss et al. [8] compared deep neural networks, gradient-boosted trees, and random forests for statistical arbitrage in US equities, finding that ensemble methods achieved daily returns of 0.33–0.45% before transaction costs over the period 2000–2015.

More recently, studies have explored whether the superior in-sample performance of complex models translates to genuine out-of-sample predictability. Gu et al. [9] conducted a comprehensive comparison of machine learning methods for measuring asset risk premia and found that tree-based methods and neural networks outperform linear models, though the magnitude of improvement varies across markets and time periods.

TABLE I
Selected Stocks and Their Sector Classification

Ticker	Sector	Observations
RELIANCE.NS	Energy/Conglomerate	2220
TCS.NS	IT Services	2220
HDFCBANK.NS	Private Banking	2220
INFY.NS	IT Services	2220
ICICIBANK.NS	Private Banking	2220
HINDUNILVR.NS	FMCG	2220
SBIN.NS	Public Sector Banking	2220
ITC.NS	FMCG/Diversified	2220
MARUTI.NS	Automobile	2220
SUNPHARMA.NS	Pharmaceuticals	2220

C. Technical Indicators as Features

Technical indicators remain widely used as input features for machine learning models in financial prediction, despite the mixed empirical evidence for their predictive value in isolation [10]. Common categories include trend indicators (moving averages, ADX), momentum oscillators (RSI, MACD, Stochastic), volatility measures (Bollinger Bands, ATR), and volume-based indicators (OBV). The choice and combination of these indicators significantly influences model performance [11].

Nti et al. [11] conducted a systematic review of 122 studies on stock market prediction using machine learning and found that the majority employed between 5 and 20 technical indicators, with RSI, MACD, and moving averages being the most frequently used. They noted that the optimal feature set remains an open question and varies across markets and prediction horizons.

D. Research Gap

While the literature contains numerous studies on stock prediction with machine learning, several gaps motivate our work. First, many studies employ random rather than temporal train-test splits, making their results difficult to interpret from a practical standpoint. Second, formal statistical testing of performance differences between models is often absent. Third, comprehensive comparisons specifically on the Indian market using recent data (post-2020, including the COVID-19 market disruption and recovery) are scarce. Our study addresses these gaps by combining rigorous temporal evaluation methodology with statistical significance testing on recent NIFTY 50 data.

III. Methodology

A. Data Collection

We collected daily Open, High, Low, Close, and Volume (OHLCV) data for 10 stocks from the NIFTY 50 index, selected to represent diverse market sectors (Table I). Data spans the period January 1, 2016 to December 30, 2024, yielding approximately 2220 trading days per stock. Data was obtained from Yahoo Finance through the `yfinance` Python library.

The selection criteria ensured: (i) continuous trading history over the entire study period without delistings or extended

TABLE II
FEATURE CATEGORIES AND COUNT

Category	Count	Representative Features
Returns	4	1d, 5d, 10d, 20d returns
Moving Averages	4	Price relative to SMA(5,10,20,50)
Volatility	6	Rolling std, BB width/position, ATR
Momentum	9	RSI, MACD (3), Stochastic (2), Williams %R, CCI
Volume & Trend	5	Volume ratio, OBV change, ADX (3)
Candlestick	3	Body size, upper/lower shadow
Total	**31**	

trading halts; (ii) representation of at least six distinct sectors to assess cross-sector generalizability; and (iii) inclusion of stocks with varying market capitalization and liquidity profiles within the large-cap space.

B. Feature Engineering

We computed 31 technical indicator features from the raw OHLCV data, organized into five categories as described below and summarized in Table II.

1) Returns and Price Momentum: Historical returns at multiple horizons (1-day, 5-day, 10-day, and 20-day) capture short and medium-term price momentum. These are computed as simple percentage changes: $r_t^{(k)} = (P_t - P_{t-k})/P_{t-k}$.

2) Moving Average Features: We compute the relative position of the current price with respect to Simple Moving Averages (SMA) of periods 5, 10, 20, and 50 days. These features, expressed as $(\text{Close}_t/\text{SMA}_t^{(k)}) - 1$, capture whether the stock is trading above or below its recent average levels.

3) Volatility Features: Rolling standard deviation of daily returns over 5, 10, and 20-day windows measure realized volatility at different timescales. Additionally, we include Bollinger Band width (normalized by price) and the position of the current price within the bands, as well as the 14-day Average True Range (ATR).

4) Momentum Oscillators: We include the 14-day Relative Strength Index (RSI), MACD line and signal with histogram, the Stochastic Oscillator (%K and %D lines), Williams %R, and the 20-day Commodity Channel Index (CCI). These oscillators are designed to identify overbought or oversold conditions.

5) Volume and Trend Indicators: Volume features include the volume change ratio and On-Balance Volume (OBV) momentum. The Average Directional Index (ADX) with its positive and negative directional indicators captures trend strength regardless of direction.

C. Target Variable

The binary target variable is defined as:

$$y_t = \begin{cases} 1 & \text{if } P_{t+5} > P_t \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where P_t denotes the closing price at time t. We chose a 5-day prediction horizon as it represents a practical short-term trading horizon that balances signal strength against the increased uncertainty of longer horizons. The average class balance across stocks ranged from 51.04% to 56.24% positive (price increase) instances, indicating a slight upward bias consistent with the general long-term upward drift of equity markets.

D. Classification Models

We evaluate seven classification algorithms spanning different learning paradigms:

Logistic Regression (LR): A linear model that estimates class probabilities via the logistic function. We use L2 regularization with $C = 1.0$.

Support Vector Machine (SVM): We employ the SVM with a Radial Basis Function (RBF) kernel ($C = 1.0$, $\gamma =$ scale), which can model non-linear decision boundaries through the kernel trick.

Random Forest (RF): An ensemble of 200 decision trees with maximum depth of 10, using bagging and random feature subsets to reduce overfitting and variance [12].

XGBoost (XGB): A gradient boosting implementation with 200 estimators, maximum depth 5, learning rate 0.05, and column subsampling ratio 0.8 [7].

K-Nearest Neighbors (KNN): An instance-based learner using $k = 15$ neighbors with distance-weighted voting.

Gaussian Naive Bayes (NB): A probabilistic classifier assuming feature independence and Gaussian-distributed features within each class.

Gradient Boosting (GB): The scikit-learn implementation with 150 estimators, maximum depth 4, learning rate 0.05, and row subsampling ratio 0.8.

E. Experimental Design

1) Temporal Train-Test Split: To avoid look-ahead bias—a critical concern in financial prediction that arises when future information leaks into the training set—we employ a strictly chronological 80/20 split. The first 80% of observations (approximately 1737 samples per stock, covering roughly 2016–2023) constitute the training set, and the remaining 20% (approximately 434 samples, covering 2023–2024) form the test set. No shuffling is performed.

2) Feature Standardization: All features are standardized using z-score normalization ($z = (x - \mu)/\sigma$) where the mean μ and standard deviation σ are computed exclusively on the training set and applied to both training and test sets. This prevents information leakage from the test period.

3) Evaluation Metrics: We report four standard classification metrics:

- **Accuracy:** Proportion of correctly classified instances.
- **Precision:** Proportion of predicted positive instances that are truly positive.
- **Recall:** Proportion of actual positive instances that are correctly identified.
- **F1-Score:** Harmonic mean of precision and recall.
- **AUC-ROC:** Area under the Receiver Operating Characteristic curve, measuring discrimination ability independent of the classification threshold.

TABLE III
AGGREGATE MODEL PERFORMANCE (MEAN ± STD ACROSS 10 STOCKS)

Model	Accuracy	F1-Score	AUC-ROC
Random Forest	.533 ± .031	.558 ± .092	.527 ± .032
XGBoost	.522 ± .027	.520 ± .082	.537 ± .032
Gradient Boosting	.518 ± .023	.523 ± .082	.524 ± .026
KNN	.515 ± .023	.547 ± .041	.522 ± .027
SVM (RBF)	.513 ± .031	.516 ± .083	.527 ± .036
Logistic Regression	.510 ± .036	.491 ± .104	.534 ± .041
Naive Bayes	.472 ± .052	.214 ± .200	.514 ± .038

4) Statistical Significance Testing: We employ the Friedman test [13] as an omnibus non-parametric test to determine whether significant differences exist among the seven models when evaluated across 10 stocks (treated as repeated measures). When the Friedman test is significant, we perform post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction to identify specific pairs that differ significantly. Cohen's d is computed to quantify effect sizes.

IV. RESULTS

A. Aggregate Model Performance

Table III presents the mean and standard deviation of performance metrics aggregated across all 10 stocks. Random Forest achieved the highest mean accuracy (53.29%), followed by XGBoost (52.23%) and Gradient Boosting (51.84%). All models except Naive Bayes achieved mean accuracy above the 50% random baseline, though the margins are modest.

In terms of F1-Score, Random Forest again leads (0.558), with KNN showing notably consistent performance (0.547 with the lowest standard deviation of 0.041). Naive Bayes performs poorly across all metrics, particularly in F1-Score (0.214), suggesting that the independence assumption is severely violated for technical indicator features.

Interestingly, for AUC-ROC—which measures ranking ability independent of the decision threshold—XGBoost achieves the highest value (0.537), slightly outperforming Random Forest (0.527). Logistic Regression also performs competitively on this metric (0.534), suggesting that while its hard classification decisions are suboptimal, its probability estimates maintain reasonable discriminative power.

B. Per-Stock Analysis

Fig. 1 shows the accuracy of each model across individual stocks. Performance varies considerably across stocks: INFY.NS and TCS.NS (IT sector) generally yielded higher prediction accuracy (up to 58.4% for Random Forest on INFY), while SUNPHARMA.NS and HINDUNILVR.NS proved more difficult to predict, with several models falling below the 50% baseline.

Fig. 2 presents a heatmap of F1-Scores across all model-stock combinations. The highest individual F1-Score observed was 0.719 (Naive Bayes on SBIN.NS), though this reflects a degenerate case where the model predicts the majority class almost exclusively. Among well-calibrated models, Random Forest on INFY.NS achieved 0.667.

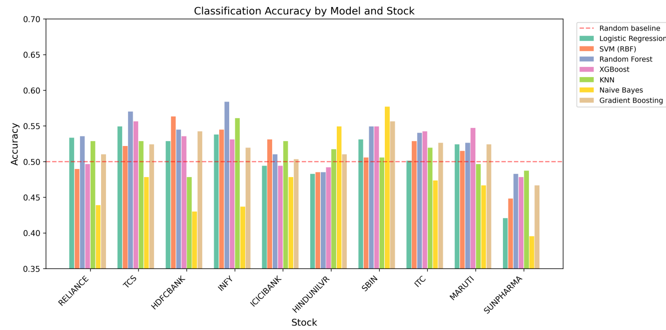

Fig. 1. Classification accuracy by model across 10 NIFTY 50 stocks. The dashed red line indicates the 50% random baseline. Ensemble methods (Random Forest, XGBoost, Gradient Boosting) generally outperform other approaches, though stock-specific variation is substantial.

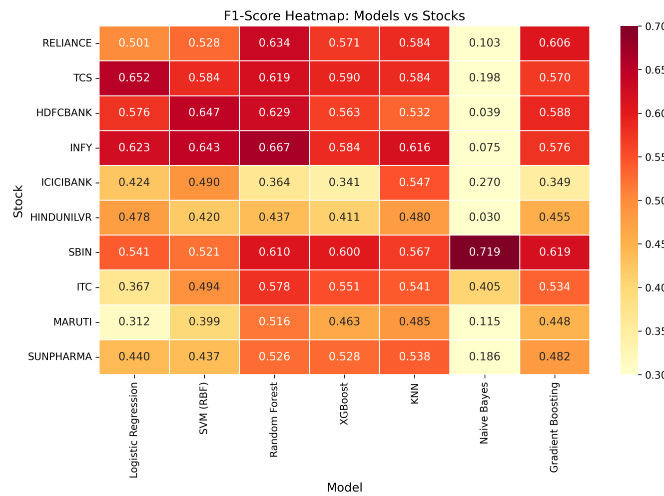

Fig. 2. F1-Score heatmap across models and stocks. Darker cells indicate higher F1-Scores. Random Forest and Gradient Boosting show the most consistent performance across stocks. Naive Bayes exhibits extremely low F1-Scores on most stocks due to degenerate predictions.

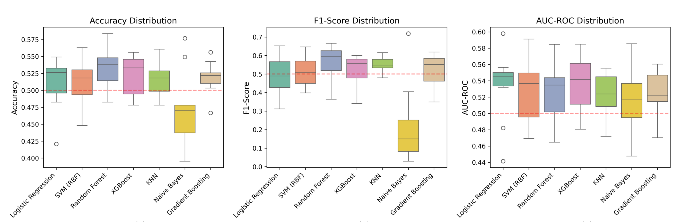

Fig. 3. Box plots showing the distribution of metrics across 10 stocks for each model. The red dashed line represents the 0.5 baseline. Random Forest shows the highest median with moderate variance; KNN demonstrates the most consistent F1-Scores.

C. Distribution of Metrics

Fig. 3 displays box plots showing the distribution of accuracy, F1-Score, and AUC-ROC across the 10 stocks for each model. Random Forest exhibits both the highest median accuracy and a relatively tight interquartile range, suggesting consistent performance. KNN shows the smallest variance in F1-Score, while XGBoost demonstrates the best AUC-ROC distribution.

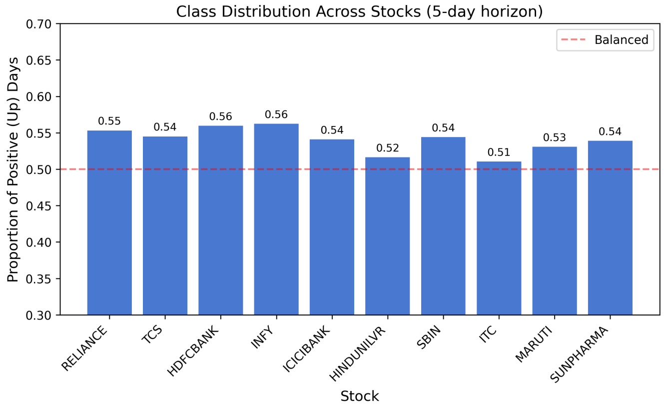

Fig. 4. Class distribution showing the proportion of positive (upward) 5-day movements for each stock. The slight positive bias (51–56%) reflects the long-term upward drift of equity markets over the study period.

TABLE IV
SELECTED PAIRWISE WILCOXON TESTS (BONFERRONI CORRECTED)

Model A	**Model B**	**p-value**	**Sig.**
Random Forest	Logistic Reg.	0.0020	Yes
Random Forest	Naive Bayes	0.0195	No
XGBoost	Naive Bayes	0.0195	No
Random Forest	XGBoost	0.2031	No
Random Forest	Grad. Boosting	0.1035	No
XGBoost	KNN	0.6250	No

D. Class Distribution

Fig. 4 shows the proportion of positive (upward movement) days for each stock. The class balance ranges from 51.04% (ITC) to 56.24% (INFY), indicating a mild positive bias across the study period. This imbalance is modest but should be considered when interpreting accuracy figures—a naive majority-class classifier would achieve 51–56% accuracy depending on the stock.

E. Statistical Significance Analysis

1) Friedman Test: The Friedman test, treating stocks as blocks and models as treatments, yielded $\chi^2 = 15.36$ with $p = 0.018$, indicating statistically significant differences among the seven models at the conventional $\alpha = 0.05$ level.

2) Post-Hoc Pairwise Comparisons: Table IV presents selected results from pairwise Wilcoxon signed-rank tests with Bonferroni correction (adjusted $\alpha = 0.05/21 = 0.00238$). After correction, only one pairwise comparison reaches significance: Random Forest vs. Logistic Regression ($W = 0, p = 0.002$), confirming that Random Forest's superiority over linear classification is robust.

3) Effect Sizes: Cohen's d analysis reveals that the difference between Random Forest and Logistic Regression has a large effect size ($d = 1.17$), while the difference between Random Forest and XGBoost is medium ($d = 0.51$). Most other pairwise comparisons yield small or negligible effect sizes, suggesting that the practical difference among the middle-ranking models is minimal.

TABLE V
MODEL RANKINGS BY ACCURACY AND MEAN RANK

Rank	**Model**	**Mean Acc.**	**Mean Rank**
1	Random Forest	0.5329	2.30
2	XGBoost	0.5223	3.40
3	Gradient Boosting	0.5184	3.85
4	KNN	0.5152	3.75
5	SVM (RBF)	0.5133	4.20
6	Logistic Regression	0.5103	4.70
7	Naive Bayes	0.4724	5.80

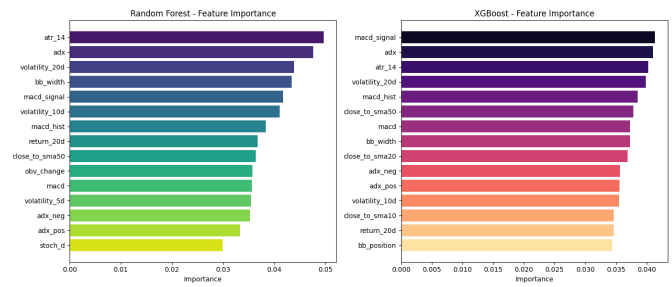

Fig. 5. Top 15 features ranked by importance for Random Forest (left) and XGBoost (right). Both models consistently rank trend strength (ADX), volatility measures (ATR, BB width), and MACD components among the most predictive features.

4) Model Rankings: Table V presents the final model rankings based on both mean accuracy and mean Friedman rank across stocks.

F. Feature Importance Analysis

Fig. 5 presents the feature importance rankings from Random Forest (Gini importance) and XGBoost (gain-based importance), averaged across all 10 stocks.

The top five most important features, averaged across both models, are:

1) Average True Range, 14-day (ATR_{14}): 0.0450
2) Average Directional Index (ADX): 0.0443
3) 20-day rolling volatility: 0.0419
4) MACD signal line: 0.0416
5) Bollinger Band width: 0.0404

Notable observations include: (i) volatility and trend-strength features dominate over pure momentum oscillators; (ii) longer-horizon features (20-day volatility, price relative to SMA-50) rank higher than short-term equivalents; (iii) volume-based features (OBV change, volume ratio) rank in the lower half, suggesting that volume information provides limited incremental value beyond what price-based features capture; and (iv) simple candlestick features (body size, shadow ratios) rank lowest, indicating limited predictive value for multi-day horizons.

Fig. 6 shows the correlation structure among the top 15 features. High correlations exist between MACD components and between volatility measures at different timescales, suggesting some redundancy in the feature set that could be addressed through dimensionality reduction.

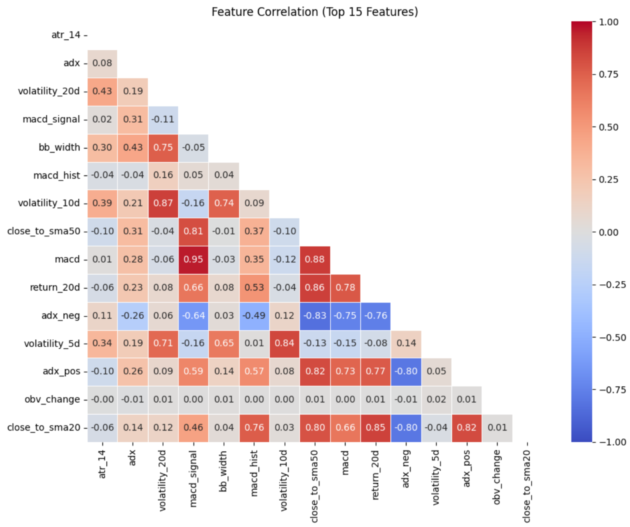

Fig. 6. Correlation heatmap of the top 15 features. Strong positive correlations exist within the MACD family and among volatility measures. The ADX-related features show moderate correlation with volatility but low correlation with momentum oscillators.

V. Discussion

A. Interpretation of Results

The overall accuracy levels observed in this study (51–53% for the best models) are modest but consistent with findings in the broader literature on stock prediction using publicly available technical indicators [6], [9]. Several interpretations are possible:

Market efficiency perspective: The results are broadly consistent with the weak-form efficient market hypothesis, which predicts that past price and volume information should not enable systematic above-chance prediction. The 2–3 percentage point improvement over random guessing achieved by the best models, while statistically significant, may not survive transaction costs in practical trading scenarios.

Signal-to-noise ratio: Financial time series are characterized by extremely low signal-to-noise ratios. Even sophisticated non-linear models struggle to extract predictive patterns when the underlying signal is weak relative to market noise [14].

Feature limitations: Our feature set is restricted to price and volume-derived technical indicators. The exclusion of fundamental data (earnings, valuations), macroeconomic indicators, sentiment data, and cross-asset information likely limits predictive power. Modern factor models suggest that a combination of technical and fundamental factors provides richer predictive signals [9].

B. Model Comparison Insights

The superior performance of Random Forest aligns with its known strengths: robustness to noise through ensemble averaging, inherent feature selection through random subspace sampling, and ability to capture non-linear interactions without extensive hyperparameter tuning. The close performance of XGBoost and Gradient Boosting further supports the hypothesis that ensemble tree methods are well-suited to tabular financial data.

The relatively poor performance of Naive Bayes is unsurprising given the strong inter-feature correlations (Fig. 6), which violate the conditional independence assumption. Its extremely low F1-Scores on most stocks indicate degenerate predictions where the model overwhelmingly predicts one class.

Interestingly, Logistic Regression achieves competitive AUC-ROC (0.534) despite lower accuracy, suggesting that its probability estimates preserve reasonable ranking ability. This has implications for applications where probability calibration matters more than hard classification.

C. Sector-Dependent Predictability

Our results reveal meaningful variation in predictability across sectors. IT sector stocks (TCS, INFY) were generally more predictable, while pharmaceutical (Sun Pharma) and FMCG (Hindustan Unilever) stocks proved harder to forecast. This may relate to differences in: (i) institutional vs. retail participation; (ii) response patterns to market-wide vs. sector-specific information; or (iii) the degree to which technical trading patterns are present in different sectors.

D. Feature Importance Implications

The dominance of volatility and trend-strength features over momentum oscillators suggests that information about *market regime* (trending vs. mean-reverting, high vs. low volatility) is more predictive of future direction than the specific level of oscillators like RSI or Stochastic. This finding has practical implications for feature engineering: rather than including many redundant momentum oscillators, practitioners might benefit from focusing on regime-identification features.

The relatively low importance of short-term features (1-day returns, 5-day SMA proximity) compared to longer-term features (20-day volatility, 50-day SMA proximity) is consistent with the 5-day prediction horizon. Features computed at timescales aligned with or longer than the prediction horizon capture the relevant market context more effectively.

E. Limitations

This study has several limitations that should be acknowledged:

1) **Single split evaluation:** While our temporal split avoids look-ahead bias, using a single split point makes results sensitive to the specific market conditions in the test period (2023–2024). Walk-forward validation with multiple windows would provide more robust estimates.
2) **Fixed hyperparameters:** We used reasonable default hyperparameters without extensive tuning. Cross-validated hyperparameter optimization might improve individual model performance, though it risks overfitting to the training period.
3) **Feature set scope:** We limit features to technical indicators derivable from OHLCV data. Incorporating alternative data sources (news sentiment, options-implied volatility, cross-asset correlations) might improve predictions.

4) **Transaction costs:** We do not account for trading costs, slippage, or market impact, which would further erode the economic significance of the modest accuracy improvements observed.
5) **Survivorship bias:** By selecting stocks that remained in the NIFTY 50 throughout the study period, we exclude stocks that were delisted or removed from the index, potentially introducing survivorship bias.

VI. Conclusion

This study presents a rigorous comparative evaluation of seven machine learning classifiers for short-term (5-day) stock price direction prediction on the Indian equity market using publicly available technical indicators. Our findings can be summarized as follows:

1) Ensemble tree-based methods, particularly Random Forest (53.29% accuracy) and XGBoost (52.23%), consistently outperform linear models, kernel methods, and instance-based approaches, though the margin of improvement is modest.
2) The Friedman test confirms statistically significant differences among models ($p = 0.018$), with Random Forest vs. Logistic Regression being the only pair reaching significance after Bonferroni correction.
3) Feature importance analysis reveals that volatility measures (ATR, Bollinger Band width) and trend strength indicators (ADX) are more predictive than traditional momentum oscillators, suggesting that market regime identification is key.
4) Despite statistical significance, the practical margin over random guessing (2–3%) is small, supporting the view that public technical indicators alone provide limited predictive power for short-term price movements.

These results have implications for both researchers and practitioners. For researchers, they underscore the importance of proper experimental methodology (temporal splits, statistical testing) and tempered expectations when working with public market data. For practitioners, they suggest that while machine learning can extract marginal signals from technical indicators, robust trading strategies likely require additional information sources and careful consideration of transaction costs.

Future work could extend this study in several directions: (i) incorporating fundamental and alternative data features; (ii) employing walk-forward validation with expanding or sliding windows; (iii) evaluating deep learning architectures (LSTM, Transformer) that can model temporal dependencies directly; (iv) assessing economic significance through simulated trading strategies net of costs; and (v) investigating regime-dependent model performance to determine whether predictability varies across bull, bear, and sideways markets.

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